

Course Information

Course Number: VIST 370 & VIZA 689
Course Title: Interactive Virtual Environment
Time: Tuesday/Thursday 12:45 to 2:00 pm
Location: LAAH 262
Credit Hours: 3

Instructor Details

Lead instructor: Jinsil Hwaryoung Seo
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Office Hours: By appointment

Instructor: Michael Bruner
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Office Hours: By appointment (MW in person)

Teaching Assistant: John Alberse
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Office Hours: By appointment

Course Description

Languages and techniques useful for the creation of real time virtual environments; definition of formal scene description structures; modeling and transformation techniques; simulation techniques; behaviors and message passing; user interaction and animation; multi-user environments; creating virtual interfaces; scripting techniques.

Prerequisites: VIST 271

*VIST 370 introduces the student to the three major focus areas in mixed/augmented/virtual environments: interactive 360 video, interactive projection & virtual reality. Students will learn advanced visualization techniques in the interactive/mixed reality environments focusing on **interaction and immersion**. The course aims to provide an overview of mixed reality with topics including input/output devices, 3D interaction techniques, design guidelines, and user study methods. Students will gain hands-on experience designing mixed reality experiences creating applications and demonstrations focusing on raising awareness of challenges in particular demographics, underserved groups, specific user groups, local communities. Students are expected to have some experience with computer programming and 3D graphics tools prior to taking this course.*

Special Course Designation

This is a stacked course which means it has both undergraduate and graduate students.

Course Learning Outcomes

At the conclusion of VIST 370/VIZA689, the student will be expected to:

- Understand foundational concepts in mixed reality
- Experience state of the art mixed reality applications
- Define a methodology for managing small group collaboration.
- Employ a variety of media based on the creation of a simulation.
- Construct accurate simulation models based on physically based processes.
- Assemble GUI's based on a variety of interactive techniques and sensors.
- Explore embodied interaction in mixed media environments
- Apply 3D interaction techniques to the design and development of mixed reality environments.
- Produce team based immersive virtual reality applications.
- Critically evaluate the design, purpose, benefits, and limitations of different mixed reality applications.

At the conclusion of VIZA689, the student will be expected to:

- Produce submission materials for competitive events including conferences, competitions, and exhibitions.

Textbook / Resource Materials

The course doesn't require a textbook. However we have reference materials and required readings. They will be electronically available on the Learning Management System Canvas and/or the class Google Drive. Online resources, which include lecture notes and class presentations, reading assignments, and other references, are for your use in this course and to assist your class activities. This information is not all-inclusive of the course content and does not replace attendance and participation in class.

Hardware / Software

Hardware:

360 Camera

Arduino, webcam, projector

Head mounted display (Oculus Quest 2)

* Students will be able to check out hardware from the school.

Software:

Adobe Premiere or equivalent

Arduino IDE

TouchDesigner

Madmapper

Unity or Unreal

* Students will use student versions or free trials of the software on their computer or school computers in VIRL/Igloo room.

Course Structure

The class consists of lectures, tutorials, presentations, critique, and possibly other things (i.e. visits to the Wright Gallery for openings, guest lectures, guest performances, etc.). The class explores the three major modules in mixed/augmented/virtual environments.

- First module will be focused on spatial interaction. Students will develop interactive 360 video projects for the Igloo environment learning foundational concepts of interaction design and immersion (W1-W5)
- The second module will focus on augmented interaction in a projected environment. Students will learn physical computing and projection mapping foundations (W5-W10).
- The third module will focus on virtuality. Students will learn fundamental concepts in VR and experiment interactive techniques in Immersive VR (W10-W15).

Evaluation

Course grades will be calculated based on a combination of weighted scores for projects, homework, and exams. For the semester, the total grade is based on 100 points, which is equal to 100% of your grade. Assessment criteria for group projects will include individual technical and artistic contributions, participation, peer assessment, and overall project success. Assessment criteria for individual projects will be based upon technical and artistic merit relative to the given criteria. The instructor's qualitative judgment of the student's exercises, projects, and research will include such factors as preparation, conceptualization, technical application, documentation, and aesthetic effectiveness.

- Module 1: 25%
- Module 2: 25%
- Module 3: 25%
- Exams (midterm, final): 15%
- Weekly Assignments/Participation: 10 % (for undergraduate students)
- Weekly Assignment/Participation: 5% (for graduate students)
- Conference paper + documentation video: 5% (for graduate students)
- Total 100%

*Graduate students are required to create a conference or festival-ready project with a high quality paper and a documentation video. They have to submit their projects/papers to competitive venues.

Graded Attendance and Participation

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments. Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Your grade will be based upon attendance and active participation in a wide range of class activities. You are expected to be in class from start to finish every day that we have class, (There are exceptions such as health, please see the University Attendance policy: <https://student-rules.tamu.edu/rule07/>).

It is the student's responsibility to inform the instructor of any intended absences. **A student's final grade will be dropped a letter grade for each unexcused absence after 3 unexcused absences (from 4 unexcused absences).**

Excused absences: A list of excused absences can be found in Student Rule 7.1 (<http://studentrules.tamu.edu/rule07>). Except for absences due to religious Holidays, the student must notify his or her instructor in writing (acknowledged email message is acceptable) prior to the date of absence if such notification is feasible. In cases where advance notification is not feasible (e.g. accident, or emergency) the student must provide notification by the end of the second working day after the absence. This notification should include an explanation of why notice could not be sent prior to the class. If the absence is excused, the instructor must either provide the student with an opportunity to make up any quiz, exam or other graded activities or provide a satisfactory alternative to be completed within 30 calendar days from the last day of the absence.

Excused Absences for Religious Holidays: Texas House Bill 256 (effective 9/1/03) states "An institution of higher education shall excuse a student from attending classes or other required activities, including examinations, for the observance of a religious holy day, including travel for that purpose. A student whose absence is excused under this subsection may not be penalized for that absence and shall be allowed to take an examination or complete an assignment from which the student is excused within a reasonable amount of time after the absence."

Extra Credits

Opportunities to earn extra credit are not promised, but the instructor may opt to offer supplemental assignments for extra credit. Details will be determined per assignment and must be agreed upon by both the instructor and the student.

Late Work Policy

- *Late work will have 10 points deducted for every day that it is late.*
- Work submitted by a student as makeup work for an excused absence is not considered late work and is exempted from the late work policy. (See [Student Rule 7](#).)
- Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

- Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.
- Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor. ([Student Rule 7, Section 7.4.1](#)).
- The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence. ([Student Rule 7, Section 7.4.2](#)).
- Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Grading Policy

Course grades will be calculated based on a combination of weighted scores for projects, assignments, participations, and exams. For the semester, the total grade is based on 100 points, which is equal to 100% of your grade.

Letter grades will be assigned according to the following guideline:

A = 90-100 (Excellent)

B = 80-89 (Good)

C = 70-79 (Satisfactory)

D = 60-69 (Passing)

F = 59 and below (Failing)

I=Incomplete.

To aid in understanding the method by which your project assignments are evaluated, read the following carefully. The actual grading criteria will be related to these categories but will be more specific in nature.

F: 59 and below: *The student work is unresolved; the intentions are unclear and major criteria or goals lack resolution. Presentation is incomplete and/or of poor quality. There is a complete lack of problem solving, creative content and/or technical merit.*

D: 60 -69: *The solution has problems in two or more major areas; skill and design development is marginal or incomplete. The project lacks imagination and/or design potential.*

C: 70 -79: *The student has solved the problem, but the solution lacks depth of understanding; Some program goals are not completely satisfied and the solution contains little promise even though most issues have been addressed.*

B: 80 -89: *The student work shows imagination and potential. Presentation and design content is good; program requirements are fulfilled but in need of more refinement or development. There are no major issues that would require a total redesign of the project.*

A: 90- 100: The student work has imagination and the solutions to the problems show understanding and thought. The problem solution is highly developed and well presented; the entire assignment shows depth and breadth and is well coordinated. The assignment's potential has been achieved.

Course Schedule

This below schedule is tentative and may change (notice will be given if changes are made):

Week	Tuesdays	Thursdays
1 Aug. 22 Aug. 24	Lecture: Introduction/ Course overview Mixed Reality Continuum, Immersion	Class Activity: Bio/portfolio Presentations Lecture : Introduction of interactive 360 video Module 1 project Introduction Team formation
2 Aug. 29 Aug. 31	Lecture: Interactive narratives Demo: 360 video camera/editing	Demo: Igloo+Unity+Vive Controllers Student Presentation: 360 videos research / Narrative flow
3 Sep. 5 Sep. 7	Lecture: Design for Good, Human-Centered Design Demo: Unity Intro, Source Control, File Sharing	Demo: Unity continued, Interaction and Sequencing Class Activity: Branching Narrative layout, Figma Outline
4 Sep. 12 Sep. 14	Individual team meetings - progress check	Class Activity: Testing projects in Igloo
5 Sep. 19 Sep. 21	Module 1 project presentations	Module 2 Project Introduction Lecture: Introduction of Interactive Projection Due: Module 1 project deliverables
6 Sep. 26 Sep. 28	Lecture: Interactivity & Embodiment I Demo: Arduino	Lecture: Projection Mapping DEMO: Touchdesigner, Madmapper
7 Oct. 3 Oct. 5	Lecture: Interactivity & Embodiment II Student Presentation: Project pitch presentation	Midterm Exam
8 Oct. 10 Oct. 12	No Class: Fall Break	Project progress check, individual team meetings
9 Oct. 17 Oct. 19	Installation test in VIRL	Project progress check, individual team meetings
10 Oct. 24 Oct. 26	Module 2 project presentations	Lecture: Introduction of Virtual Reality Module 3 Introduction

11 Oct. 31 Nov. 2	Team Formation Assign Oculus Quests to teams Tutorial: How to setup VR (in VIRL)	Lecture : Navigation, Manipulation, and Control I (Controllers, Sensors, Trackers)
12 Nov. 7 Nov. 9	Lecture : Navigation, Manipulation, and Control II Demo: 3D scanning	Lecture : 3D UI/UX Low-fi prototype presentation
13 Nov. 14 Nov. 16	Lecture: VR Research approaches and evaluation/ IRB	Progress Checks (VIRL)
14 Nov. 21 Nov. 23	Lecture: Documentation Video Creation Due: High-fidelity prototype	No Class
15 Nov. 28 Nov. 30	Project progress check, individual team meetings	Module 3 Project presentations Video presentation/Slides
16 Dec. 5	VIST Show - Teams have to demo the projects Due: Final project deliverables	
Final Exam: Dec 12, 2023 9am		

Optional Course Information Items

Canvas (LMS): Information on Canvas is for the sole use of Texas A&M University students enrolled in the class. This online repository contains materials (images, articles, etc.) copyrighted by others that have been included for the purpose of instruction only, under the "fair use" provision of the U.S. copyright laws. Copying these materials to any other site or use of these materials for any other purpose is prohibited. Please log in at <https://canvas.tamu.edu/>

Google Drive: Class slides and other reference materials will be shared via a class Google Team Drive. Assignment submissions will be done via Google Drive as well. You will be added to it in the first week of the semester. <https://drive.google.com/drive/u/0/folders/0AAXvPVvEvMFUk9PVA>

Slack: Slack will be utilized as a main communication channel for the class: IVE_Fall2023
 Invitation Link:
https://join.slack.com/t/ivefall2023/shared_invite/zt-21imay0gm-dAbQbljeubidCDLZo6BxPg

Miro: for brainstorming ideas. <https://miro.com/>

Padlet: for class discussion https://padlet.com/hwaryoung/ive_fall2023-xibbkjsj162p88kw

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" ([Student Rule 7, Section 7.4.1](#)).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2](#)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20](#)).

Texas A&M at College Station

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below). Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Texas A&M at College Station

Disability Resources is located in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see [University Rule 08.01.01.M1](#)):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, a person who is subjected to the alleged conduct will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

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Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with [Counseling and Psychological Services](#) (CAPS).

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's [Title IX webpage](#).

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors that influence a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care by utilizing available resources and services on your campus

Texas A&M College Station

Students who need someone to talk to can contact Counseling & Psychological Services (CAPS) or call the TAMU Helpline (979-845-2700) from 4:00 p.m. to 8:00 a.m. weekdays and 24 hours on weekends. 24-hour emergency help is also available through the National Suicide Prevention Hotline (800-273-8255) or at suicidepreventionlifeline.org.

Use of AI in the Classroom

AI tools are allowed to generate content (text, video, audio, images) for any work (assignments, activities, responses, etc). Since those will be part of your evaluation in this course, any student work submitted using AI tools should clearly indicate what work is the student's work and what part is generated by the AI. When AI is being used for class projects and/or assignments, students **MUST** cite which AI application they used, and describe how it was used in their assignment. If any part of this is confusing or uncertain, please reach out to your instructor for a conversation before submitting your work.